ECEN 3730: Board 1 Report
Astable Multivibrator

Eric Walker
9/20/26

Objective:

The objective of this PCB design project was to design, fabricate, assemble, and test a circuit using a 555 timer to generate an approximately 500 Hz signal with a 60% duty cycle. The timer output drives four LEDs through different series resistor values of 10 kΩ, 1 kΩ, 300 Ω, and 50 Ω, allowing the effect of load resistance on LED brightness to be observed. This project also served as an introduction to the complete PCB design process. The circuit was first prototyped and tested on a solderless breadboard before creating the schematic and laying out the board as a PCB in Altium. The board incorporates several good PCB design practices, including a ground plane, indicator LEDs, isolation switches, clearly labeled components, and accessible test points. Test points were included at important circuit nodes and between one LED and its series resistor so that voltage and LED current could be measured.

Plan of Record (POR):

The plan for Board 1 was to create a simple 555 timer astable oscillator circuit powered from a 5V DC source. The 555 timer is configured in astable mode with a target output frequency of approximately 500 Hz and a duty cycle of approximately 60%. The oscillator output will drive four LED branches containing different series resistor values so that effect of the load resistance on the LED brightness could be observed.

The four LED branches will use resistor values of 10 kΩ, 1 kΩ, 300 Ω, and 50 Ω. One of the branches will include a test point so that the voltage across the resistor could be measured and used to calculate current through the branch. Additionally, test points will be included on all important nodes such as the 5V power rail, and the 555 output. In addition, isolation switches will be included to disconnect the 555 timer from the input voltage as well as disconnect the LED branches from the 555 timer.

The PCB will also include a continuous ground plane, clearly labeled components, indicator LEDs, decoupling capacitors, and short organized signal routing. The circuit will first be constructed and tested on a solderless breadboard to verify the design and choose components before creating the schematic and PCB layout in Altium.

Requirements for Board to work:

- 5V indicator light turns on when plugged in from the AC to DC cord
- The 555-timer output produces a square wave with ~500 Hz frequency and ~66% duty cycle and can be measured using the test point
- The output of the 555-timer turns on all 4 LEDS at different brightness
- The voltage across the 1 kΩ resistor can be measured using the test point
- Both the 555-timer and the LED branches can be disconnected using the switches

Sketch and calculations:

$R1 = 1k\Omega$, $R2 = 1k\Omega$, $C = 1\mu F$

$t_{high} = 0.693(R_1 + R_2)C$

$= 0.693(1000+1000)(10^{-6})$

$= 0.693(2000)(10^{-6})$

$= 1.386 \times 10^{-3}$ s

$t_{high} = 1.386$ ms

$t_{low} = 0.693 R_2 C$

$= 0.693(1000)(10^{-6})$

$= 0.693(10^{-3})$s

$t_{low} = 0.693$ ms

$T = t_{high} + t_{low}$

$= 1.386ms + 0.693ms$

$T = 2.079$ ms

$S = \frac{1}{T}$

$= \frac{1}{2.079 \times 10^{-3}}$ s

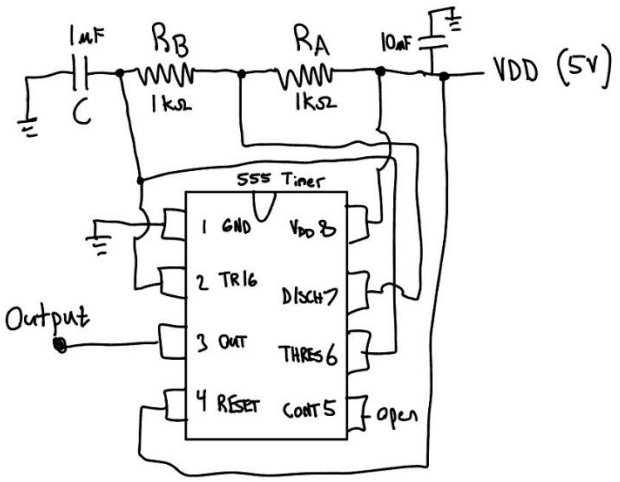
$S = 481$ Hz

$D = \frac{R_1 + R_2}{R_1 + 2R_2} \times 100\%$

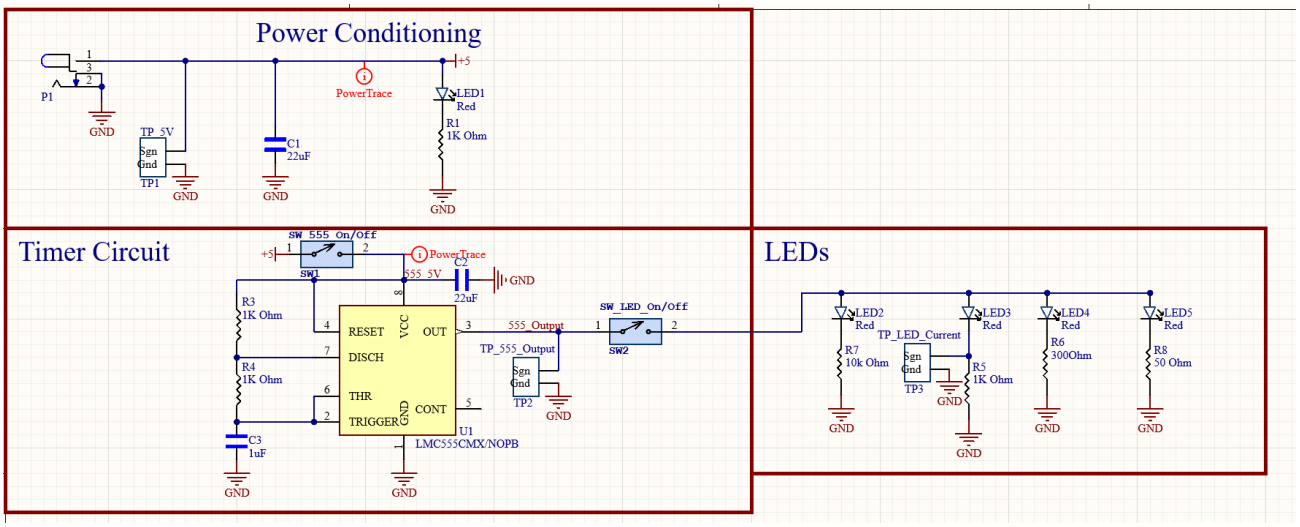
$= \frac{1000+1000}{1000+2000} \times 100\%$

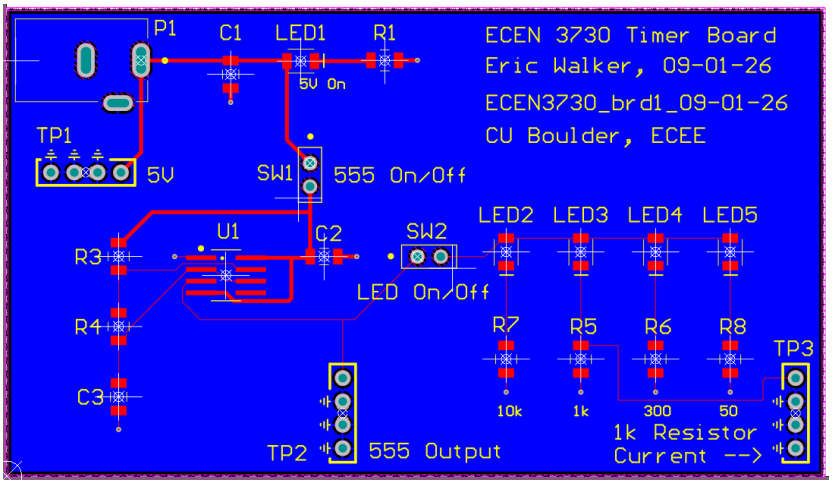
$= \frac{2}{3} \times 100\%$

$D = 66.7\%$

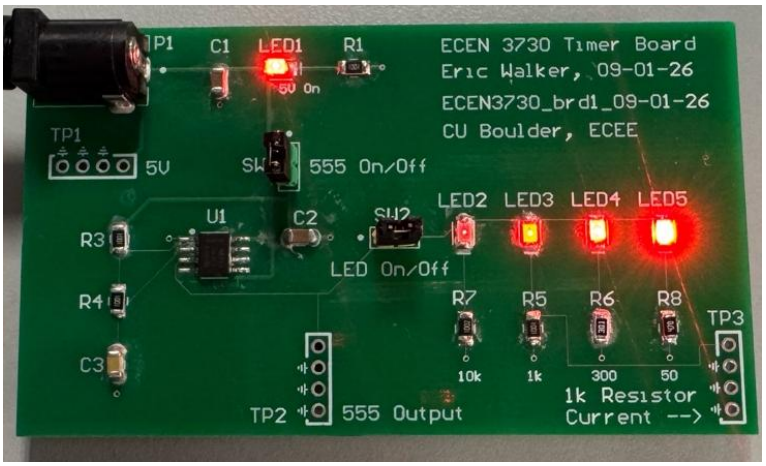


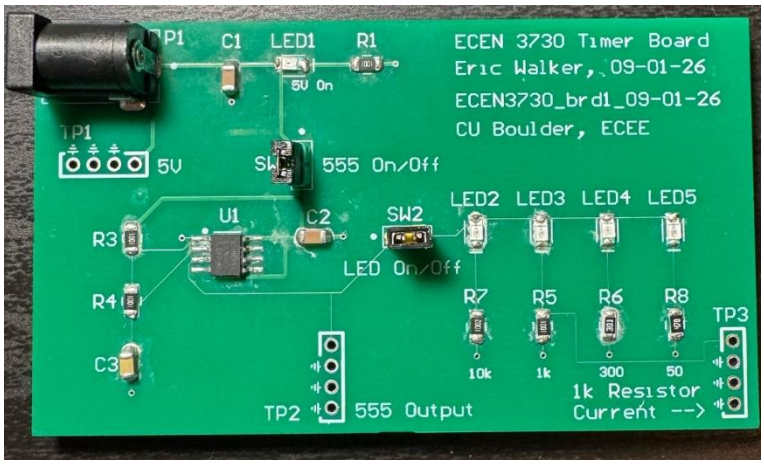
Schematic and Layout:





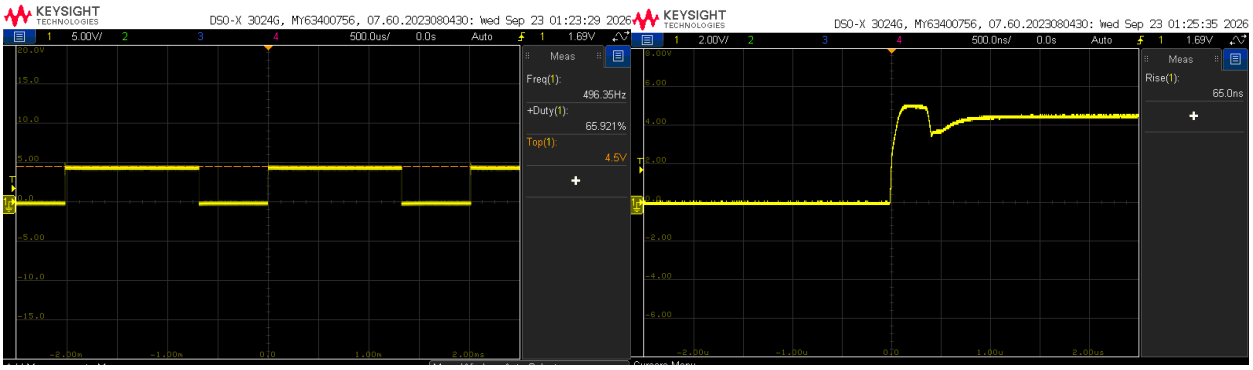
Board Implementation:





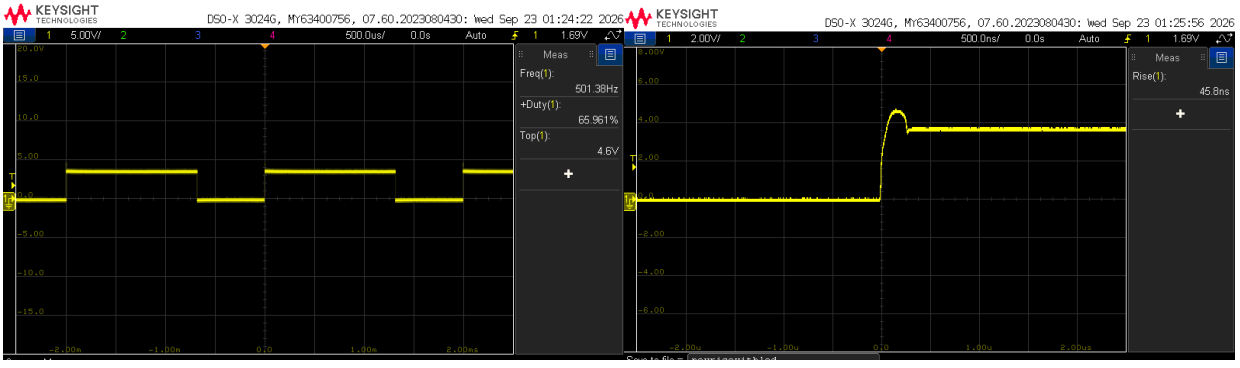
Scope Shots and Analysis:

Figure 1,2: Unloaded 555 timer output waveform and rise time



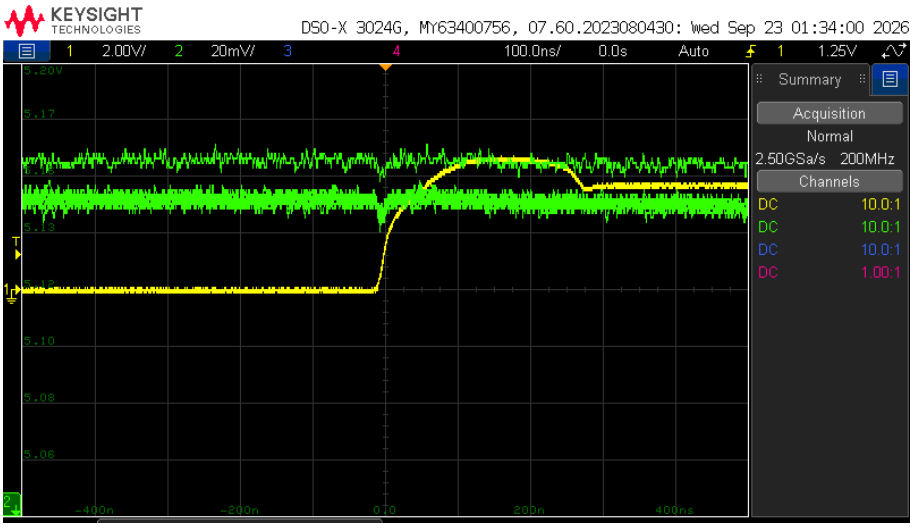
The measured output frequency was approximately 496.35 Hz with a 65.9% duty cycle and an output-high voltage of about 4.5 V. The measured rise time is approximately 65 ns.

Figure 3,4: Loaded 555 timer output waveform and rise time



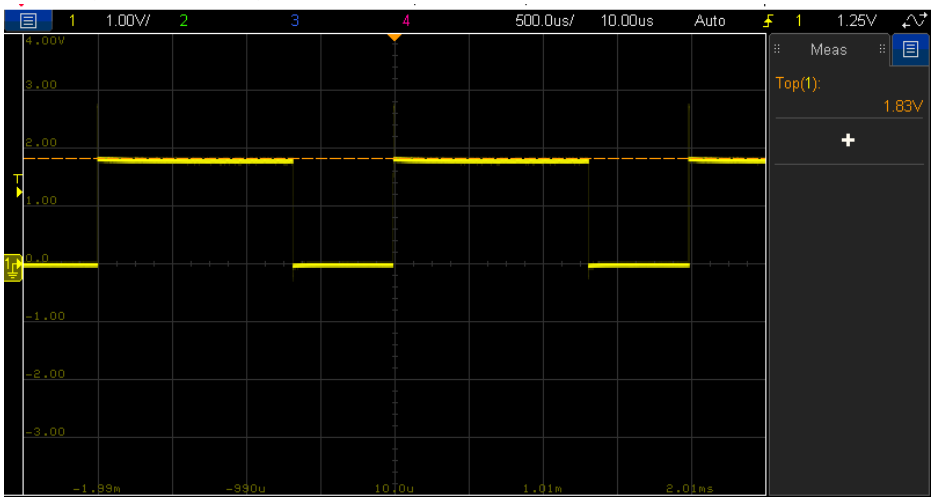
The measured output frequency was approximately 501.38 Hz with a 65.96% duty cycle and an output high voltage of 3.7V. The measured loaded rise time is approximately 45.8 ns.

Figure 5: Switching noise measured on the 5 V power rail during 555 output switching



The 5V supply rail (green) was probed while the 555-timer output was switching to observe the supply noise caused by the transient current spike. A brief disturbance was observed on the power rail at the rise of the timer's edge followed by a small settling behavior. This indicates that switching in the 555 and the load network couples noise onto the power rail. The measurement demonstrates the importance of a continuous ground plane, short power and return paths, and close decoupling capacitors which all help reduce the effects of switching noise on the power rail.

Figure 6: Voltage measured across the 1 kΩ LED series resistor



The measured voltage across the 1 kΩ resistor is 1.83V, allowing the current to be calculated through the LED. I = V/R = 1.83/1000 = 1.83 mA.

Output Analysis: (Figures 1,3)

Measurement	Predicted	Unloaded Measurement	Loaded Measurement
Frequency	480 Hz	496.35 Hz	501.38
Duty Cycle	66.7 %	65.9 %	65.96 %
Rise Time	---	65 ns	45.8 ns
Output High	---	4.5 V	3.7 V

Comparing the calculated values with the oscilloscope measurements, the measured frequency and duty cycle closely match the predicted values. The unloaded frequency was 496.35 Hz, and the loaded frequency was 501.38 Hz, compared with the predicted value of approximately 480 Hz. Similarly, the measured duty cycles of 65.9% unloaded and 65.96% loaded were close to the predicted value of 66.7%. Connecting the LED branches therefore had little effect on the timing characteristics of the 555 oscillator.

The unloaded 555 output high voltage was approximately 4.5 V, while the loaded output high voltage decreased to approximately 3.7 V. This 0.8 V reduction under load shows that the 555 output does not behave as an ideal voltage source and has finite output resistance. Although loading had little effect on frequency and duty cycle, it produced a noticeable reduction in voltage on the 555 output as it supplied current to the four LED branches.

LED Current Analysis: (Figure 6)

The voltage across each LED series resistor was measured while the 555 output was high. The corresponding branch currents were calculated using Ohms law.

Value	Voltage	Current
10 kΩ	1.90 V	0.190 mA
1 kΩ	1.83 V	1.830 mA
300 Ω	1.73 V	5.770 mA
50 Ω	1.53 V	30.60 mA

As the series resistance decreased, the LED current increased significantly. The 10 kΩ branch carried only 0.19 mA while the 50 Ω branch carried 30.6 mA. This corresponded with the observed LED brightness as the LEDs carrying greater current appeared brighter. Based on the measurements, a peak LED current of about 1–2 mA is a good minimum for an indicator LED. Since the LEDs are only on for about 70% of the cycle, this gives an average current of about 0.7–1.4 mA, which is still visibly bright while minimizing wasted power. Lower currents, such as the 0.19 mA measured in the 10 kΩ branch, may be too dim for a good indicator.

The total current calculation is:

$$I_{load} = I_{10k} + I_{1k} + I_{300} + I_{50} = 0.19 + 1.83 + 5.77 + 30.6 = 38.39\ mA = 0.03839\ A$$

To calculate the effective Thevenin output resistance:

$$R_{TH} = \frac{V_{unloaded} - V_{loaded}}{I_{load}} = \frac{0.8}{0.03839} = 20.84\ \Omega$$

Using the measured loaded and unloaded output voltages, the effective Thevenin output resistance of the 555 was estimated to be approximately 20.8 Ω. This internal output resistance explains the reduction in output voltage as the LED load current is increased.

Rise Time Analysis: (Figures 2,4)

The unloaded 555 output had a measured rise time of 65 ns, while the loaded output measured 45.8 ns. Although the loaded case produced a shorter rise time, this does not necessarily mean that the LED load caused the 555 to switch faster. The difference may be influenced by the lower loaded output amplitude as well as probe placement and inductance in the load network.

Project Analysis:

• What worked and you did well and want to do in future designs
- Creating the circuit first on a SBB and testing all aspects of the design with a scope before creating the PCB
- Organizing the schematic into separate blocks and using net labels to connect them
- Using test points, indicator LEDs, and isolation switches between functional blocks for debugging and ease of testing

• What did not work, and you will want to do differently in future designs.
- Everything on the board worked as intended and there was nothing major that caused any issues with the board.

• Were there any hard errors- why did they go wrong
- No hard errors occurred, and all functional blocks worked as expected after assembly. The only errors were soft errors related to ease of use and assembly.

• Were there any soft errors that you would like to do different next time?
- Having better labeled components would have been useful for ease of assembly and ensuring each component is placed correctly
- Including test points on each of the LED branches would have been helpful for easily measuring the voltages across the resistors and would be simple to add

Conclusion:

The 555-timer test PCB was successfully designed, assembled, and tested, with all functions working as intended. The measured frequency and duty cycle closely match the calculated values, and the LED branches demonstrated the expected relationship between resistance, current, and brightness. Testing also showed non ideal behavior such as output voltage drop under load, finite output resistance, and switching noise on power rails. Learning the full design process of prototyping on a breadboard to creating the Altium schematic and PCB layout, provided a strong introduction to PCB design, assembly, and testing practices. This project taught me and reinforced the importance of good design features during the PCB design process.